VECTORS AND LINEAR COMBINATIONS SOLUTIONS

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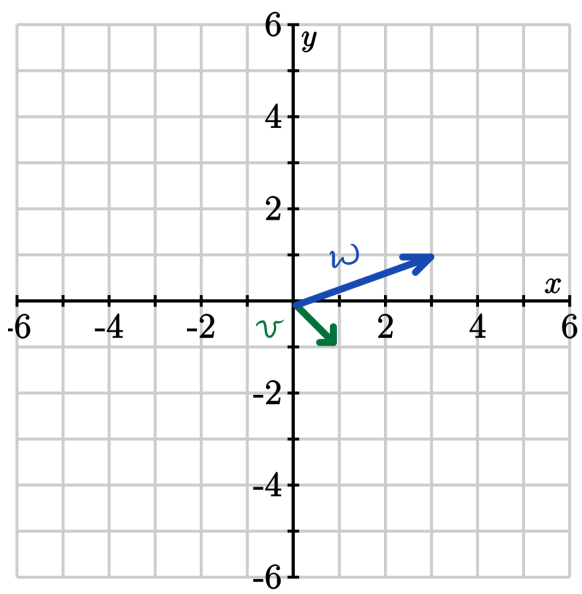
1. Consider the vectors

$$\mathbf{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \mathbf{w} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

a. Sketch these vectors below.

Solution. We view these as vectors pointing from the origin to the points $(1, -1)$ and $(3, 1)$, respectively:

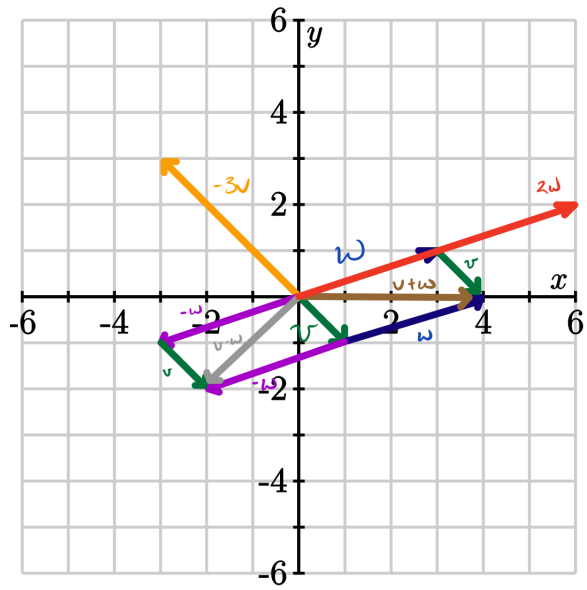


□

b. Compute the vectors $-3\mathbf{v}$, $2\mathbf{w}$, $\mathbf{v} + \mathbf{w}$, and $\mathbf{v} - \mathbf{w}$ and add them to the sketch above.

Solution.

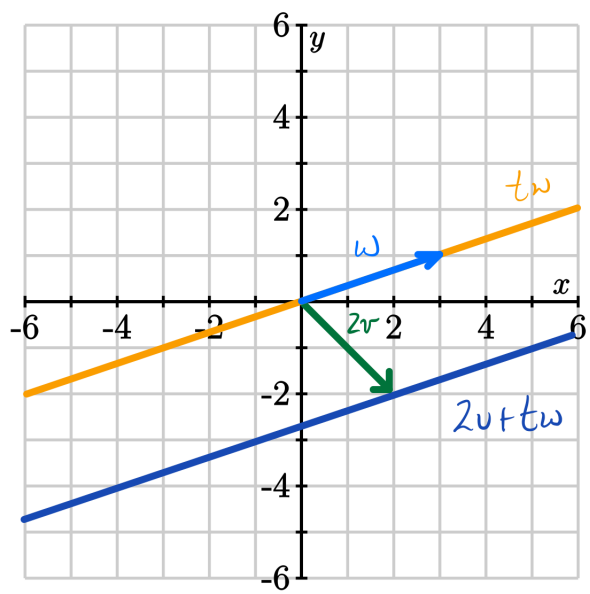
$$\begin{aligned} -3\mathbf{v} &= \begin{bmatrix} -3 \\ 3 \end{bmatrix} \\ 2\mathbf{w} &= \begin{bmatrix} 6 \\ 2 \end{bmatrix} \\ \mathbf{v} + \mathbf{w} &= \begin{bmatrix} 4 \\ 0 \end{bmatrix} \\ \mathbf{v} - \mathbf{w} &= \begin{bmatrix} -2 \\ -2 \end{bmatrix} \end{aligned}$$



□

c. Sketch below the set of vectors having the form $2\mathbf{v} + t\mathbf{w}$ where t is any scalar.

Solution. We recognize $t\mathbf{w}$ as the line with slope $1/3$ through the origin, so $2\mathbf{v} + t\mathbf{w}$ can be viewed as the line with slope $1/3$ translated by the vector $2\mathbf{v}$



We can think of the parametric curve

$$t\mathbf{w} = \begin{bmatrix} 3t \\ t \end{bmatrix}$$

in Cartesian coordinates as the line $f(x) = x/3$ and the parametric curve

$$2\mathbf{v} + t\mathbf{w} = \begin{bmatrix} 2 \\ -2 \end{bmatrix} + \begin{bmatrix} 3t \\ t \end{bmatrix} = \begin{bmatrix} 3t+2 \\ t-2 \end{bmatrix}$$

in Cartesian coordinates as the translation of f two units to the right and two units down

$$g(x) = f(x-2) + 2 = \frac{1}{3}x - \frac{8}{3}$$

Alternatively, we could also think of this simply as the line with slope $1/3$ that passes through the point $(2, -2)$.

□

d. Sketch below the line $y = 3x - 2$. Then identify two vectors $\mathbf{v}$ and $\mathbf{w}$ so that the line is described by $\mathbf{v} + t\mathbf{w}$. Are there other choices for the vectors $\mathbf{v}$ and $\mathbf{w}$?

Solution. We can represent a line with slope 3 through the origin as scalar multiples of the vector

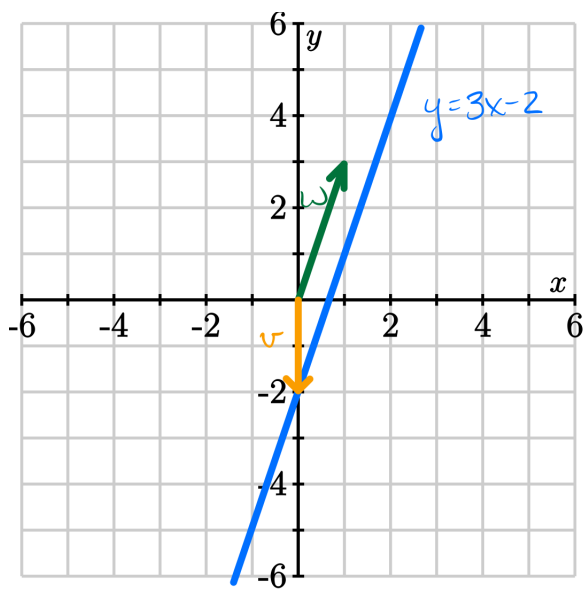
$$\mathbf{w} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

To ensure that the line passes through $(0, -2)$ we can translate by the vector

$$\mathbf{v} = \begin{bmatrix} 0 \\ -2 \end{bmatrix}$$

This yields the vector-valued function

$$f(t) = t\mathbf{w} + \mathbf{v}.$$



This representation is clearly not unique. For example, we could also represent this as the vector-valued function

$$g(t) = t\begin{bmatrix} 2 \\ 6 \end{bmatrix} + 2\begin{bmatrix} 0 \\ -1 \end{bmatrix} = 2t\mathbf{w} + 2\mathbf{v} = 2f(t)$$

□

2. Shown below are two vectors $\mathbf{v}$ and $\mathbf{w}$.

a. Express the labeled points as linear combinations of $\mathbf{v}$ and $\mathbf{w}$.

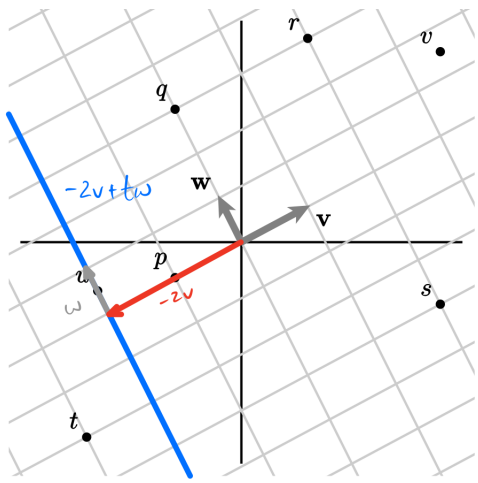
Solution.

$$\begin{aligned} p &= -\mathbf{v} \\ q &= 3\mathbf{w} \\ r &= 3\mathbf{w} + 2\mathbf{v} \\ s &= -3\mathbf{w} + 2\mathbf{v} \\ t &= -2\mathbf{w} - 3\mathbf{v} \\ u &= 1/2\mathbf{w} - 2\mathbf{v} \\ v &= 3/2\mathbf{w} + 7/2\mathbf{v} \end{aligned}$$

□

b. Sketch the line described parametrically as $-2\mathbf{v} + t\mathbf{w}$.

Solution. Translating the line$t\mathbf{w}$ through the origin by $-2\mathbf{v}$ yields



□

3. Consider the vectors

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \mathbf{v}_3 = \begin{bmatrix} -2 \\ 0 \end{bmatrix}$$

a. Find the linear combination with weights $c_1 = 2$, $c_2 = -3$, and $c_3 = 1$.

Solution.

$$2\mathbf{v}_1 - 3\mathbf{v}_2 + \mathbf{v}_3 = \begin{bmatrix} 4 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ -3 \end{bmatrix} + \begin{bmatrix} -2 \\ 0 \end{bmatrix} = \begin{bmatrix} 5 \\ -1 \end{bmatrix}$$

□

b. Can you write the vector

$$\mathbf{0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

as a linear combination of $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$? If so, describe all the ways in which you can do so.

Solution. This is trivially possible by taking the weights to be $c_1 = c_2 = c_3 = 0$.

For the non-trivial solutions, we observe that the system

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = \mathbf{b}$$

can be expressed as the augmented matrix

$$\left[\begin{array}{rrr|r} 2 & -1 & -2 & 0 \\ 1 & 1 & 0 & 0 \end{array}\right] \sim \left[\begin{array}{rrr|r} 1 & 0 & -\frac{2}{3} & 0 \\ 0 & 1 & \frac{2}{3} & 0 \end{array}\right]$$

so for any $c_3 \in \mathbb{R}$ we have

$$\frac{2}{3}c_3\mathbf{v}_1 - \frac{2}{3}c_3\mathbf{v}_2 + c_3\mathbf{v}_3 = c_3\left(\frac{2}{3}\mathbf{v}_1 - \frac{2}{3}\mathbf{v}_2 + \mathbf{v}_3\right) = \mathbf{0}.$$

That is, there are infinitely many ways to express $\mathbf{0}$ as a linear combination of $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$ □

c. Can you write the vector

$$\mathbf{0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

as a linear combination using just the first two vectors $\mathbf{v}_1$ and $\mathbf{v}_2$? If so, describe all the ways in which you can do so.

Solution. Again, yes, there is always the trivial solution $c_1 = c_2 = 0$. However, if we solve the system

$$\left[\begin{array}{rr|r} 2 & -1 & 0 \\ 1 & 1 & 0 \end{array}\right] \sim \left[\begin{array}{rr|r} 1 & 0 & 0 \\ 0 & 1 & 0 \end{array}\right]$$

then we can see that this is the *only* way. □

d. Can you write $\mathbf{v}_3$ as a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$? If so, in how many ways?

Solution. We observe from Part (b) that by taking $c_3 = 1$ we have the equation

$$\frac{2}{3}\mathbf{v}_1 - \frac{2}{3}\mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}.$$

which we solve for $\mathbf{v}_3$ to obtain the linear combination

$$\mathbf{v}_3 = -\frac{2}{3}\mathbf{v}_1 + \frac{2}{3}\mathbf{v}_2.$$

Solving the system

$$\left[\begin{array}{rr|r} 2 & -1 & -2 \\ 1 & 1 & 0 \end{array}\right] \sim \left[\begin{array}{rr|r} 1 & 0 & -\frac{2}{3} \\ 0 & 1 & \frac{2}{3} \end{array}\right]$$

we see that this is the only way. □

5. Consider the vectors

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ -1 \\ -2 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 4 \\ 4 \\ -2 \end{bmatrix}.$$

a. Can you express the vector

$$\mathbf{b} = \begin{bmatrix} 10 \\ 1 \\ -8 \end{bmatrix}$$

as a linear combination of $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$? If so, describe all the ways which you can do so.

Solution. Solving the system

$$\left[\begin{array}{rrr|r} 2 & 0 & 4 & 10 \\ -1 & 3 & 4 & 1 \\ -2 & 1 & -2 & -8 \end{array}\right] \sim \left[\begin{array}{rrr|r} 1 & 0 & 2 & 5 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

we see that there are infinitely many ways to express $\mathbf{b}$ as a linear combination of $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$:

$$\mathbf{b} = (5 - 2c_3)\mathbf{v}_1 + (2 - 2c_3)\mathbf{v}_2 + c_3\mathbf{v}_3$$

□

b. Can you express the vector

$$\mathbf{b} = \begin{bmatrix} 3 \\ 7 \\ 1 \end{bmatrix}$$

as a linear combination of $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$? If so, describe all the ways in which you can do so.

Solution. Solving the system

$$\left[\begin{array}{rrr|r} 2 & 0 & 4 & 3 \\ -1 & 3 & 4 & 7 \\ -2 & 1 & -2 & 1 \end{array}\right] \sim \left[\begin{array}{rrr|r} 1 & 0 & 2 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

we see there is no way to express $\mathbf{b}$ as a linear combination of $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$. □

c. Show that $\mathbf{v}_3$ can be written as a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$.

Solution. Solving the system

$$\left[\begin{array}{rr|r} 2 & 0 & 4 \\ -1 & 3 & 4 \\ -2 & 1 & -2 \end{array}\right] \sim \left[\begin{array}{rr|r} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{array}\right]$$

we see that

$$\mathbf{v}_3 = 2\mathbf{v}_1 + 2\mathbf{v}_2.$$

□

d. Explain why any linear combination of $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$

$$a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$$

can be rewritten as a linear combination of just $\mathbf{v}_1$ and $\mathbf{v}_2$.

Solution. Substituting the expression from Part (c) we have

$$a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3 = a\mathbf{v}_1 + b\mathbf{v}_2 + c(2\mathbf{v}_1 + 2\mathbf{v}_2) = (a + 2c)\mathbf{v}_1 + (b + 2c)\mathbf{v}_2.$$

□

6. Consider the vectors

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}.$$

For what values of k, if any, can the vector

$$\begin{bmatrix} k \\ -2 \\ 5 \end{bmatrix}$$

be written as a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$?

Solution. Applying the row operations

1. $R_1 \leftrightarrow R_3$
2. $R_2 + R_1$
3. $1/3R_2$
4. $R_3 - 3R_1$
5. $R_3 + 5R_2$
6. $R_1 - 2R_1$

we obtain the row equivalent matrix

$$\left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & k-10 \end{array}\right]$$

which we see is consistent if and only if $k = 10$. □

Remark. There are two ways to interpret this problem. The first, as above, is asking when we can express a vector as a linear combination of another two:

$$x \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix} + y \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3x+y \\ -x+y \\ x+2y \end{bmatrix} = \begin{bmatrix} k \\ -2 \\ 5 \end{bmatrix}$$

We can also think about this equation component by component, and rephrase this as asking when the three lines

$$3x + y = k, \ -x + y = -2, \text{ and } x + 2y = 5$$

all intersect in a single point. We can find the point of intersection of the latter two lines by solving the linear equation below for x

$$2x - 4 = -x + 5 \iff 3x = 9 \iff x = 3$$

Plugging $x = 3$ into into either equation yields the intersection point $(3, 1)$. Rephrasing consistency of the system as being equivalent to the line $3x + y = k$ also passing through $(3, 1)$, it's clear that there can only be one solution: the line with slope -3 through $(3, 1)$.

Since k represents the y-intercept of the third line, we find the value of k either by rewriting the point-slope form

$$y - 1 = -3(x - 3)$$

in slope-intercept form

$$y = -3x + 9 + 1 = -3x + 10$$

or by plugging $x = 3$ and $y = 1$ into the equation

$$k = 3(3) + 1 = 10.$$

7. Provide a justification for your response to the following statements or questions.

a. True or False: Given two vectors $\mathbf{v}$ and $\mathbf{w}$, the vector $2\mathbf{v}$ is a linear combination of $\mathbf{v}$ and $\mathbf{w}$

Solution. Trivially, this is true:

$$2\mathbf{v} = 2\mathbf{v} + 0\mathbf{w}.$$

□

b. True or False: Suppose $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ is a collection of m-dimensional vectors and that the matrix $\begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \ldots & \mathbf{v}_n \end{bmatrix}$ has a pivot position in every row. If $\mathbf{b}$ is any m-dimensional vector, then $\mathbf{b}$ can be rewritten as linear combination of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$.

Solution. True.

We can represent the system

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots + c_n\mathbf{v}_n = \mathbf{b}$$

by the augmented matrix $\left[\begin{array}{cccc|c} \mathbf{v}_1 & \mathbf{v}_2 & \ldots & \mathbf{v}_n & \mathbf{b} \end{array}\right]$ which, by assumption, must be row equivalent to

$$\left[\begin{array}{ccccccc|c} 1 & 0 & \ldots & 0 & \alpha_{1,m+1} & \ldots & \alpha_{1,n} & \beta_1 \\ 0 & 1 & \ldots & 0 & \alpha_{2,m+1} & \ldots & \alpha_{2,n} & \beta_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \ldots & 1 & \alpha_{m,m+1} & \ldots & \alpha_{m,n} & \beta_m \end{array}\right]$$

for some $\{\alpha_{i,j}\}_{i=1,j=m+1}^{m,n}, \{\beta_i\}_{i=1}^{m} \subset \mathbb{R}$.

This means that the coefficients $c_{m+1}, c_{m+2}, \ldots, c_n$ are free and any choice of $c_1, \ldots, c_n$ is *not* unique whenever $m < n$. If we let

$$\alpha_i = \alpha_{i,m+1}c_{m+1} + \alpha_{i,m+2}c_{m+2} + \ldots + \alpha_{i,n}c_n = \sum_{j=m+1}^{n} \alpha_{i,j}c_j, \quad 1 \le i \le m$$

then we can write

$$c_i = \beta_i - \alpha_i, \quad 1 \le i \le m.$$

As a particular solution, we could always choose $c_{m+1} = c_{m+2} = \ldots = c_n = 0$ so that

$$\mathbf{b} = \beta_1\mathbf{v}_1 + \beta_2\mathbf{v}_2 + \ldots + \beta_m\mathbf{v}_m.$$

□

c. True or False: Suppose $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ is a collection of m-dimensional vectors and that the matrix $\begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \ldots & \mathbf{v}_n \end{bmatrix}$ has a pivot position in every row and every column. If $\mathbf{b}$ is any m-dimensional vector, then $\mathbf{b}$ can be written as a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ in exactly one way.

Solution. True. This follows from Part (b): if there are pivot positions in every row and column, then the number of rows and columns must be the same; that is to say, $m = n$.

In the solution above, we observe that this means the columns containing the $\alpha_{i,j}$ don't exist, so the reduced row echelon form must be

$$\left[\begin{array}{cccc|c} 1 & 0 & \dots & 0 & c_1 \\ 0 & 1 & \dots & 0 & c_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & c_m \end{array}\right]$$

for some $(c_1, c_2, \dots, c_m) \in \mathbb{R}^m$. Therefore we have the unique solution

$$\mathbf{b} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_m\mathbf{v}_m.$$

□

d. True or False: It is possible to find two 3-dimensional vectors $\mathbf{v}$ and $\mathbf{w}$ such that every 3-dimensional vector can be written as a linear combination of $\mathbf{v}$ and $\mathbf{w}$.

Solution. False.

For any two vectors,

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}$$

we can always find a third vector, $\mathbf{b}$, by finding any non-trivial solution to the homogeneous system

$$\left[\begin{array}{ccc|c} v_1 & v_2 & v_3 & 0 \\ w_1 & w_2 & w_3 & 0 \end{array}\right]$$

We are guaranteed to have such a solution because the system is always consistent and there must always be at least one free variable. We claim that $\mathbf{b}$ cannot be expressed as a linear combination of $\mathbf{v}$ and $\mathbf{w}$.

We observe that if we form the augmented matrix

$$\left[\begin{array}{cc|c} v_1 & w_1 & b_1 \\ v_2 & w_2 & b_2 \\ v_3 & w_3 & b_3 \end{array}\right]$$

and apply the row operations
1. b_3R_3,
2. $R_3 + b_1R_1$, and
3. $R_3 + b_2R_2$

then we obtain the row-equivalent matrix

$$\left[\begin{array}{cc|c} v_1 & w_1 & b_1 \\ v_2 & w_2 & b_2 \\ b_1v_1 + b_2v_2 + b_3v_3 & b_1w_1 + b_2w_2 + b_3w_3 & b_1^2 + b_2^2 + b_3^2 \end{array}\right]$$

Since $\mathbf{b}$ was chosen to be a non-trivial solution to the system

$$\begin{aligned} b_1v_1 + b_2v_2 + b_3v_3 &= 0 \\ b_1w_1 + b_2w_2 + b_3w_3 &= 0 \end{aligned}$$

we know that

$$b_1^2 + b_2^2 + b_3^2 \neq 0$$

and hence the system

$$\left[\begin{array}{cc|c} v_1 & w_1 & b_1 \\ v_2 & w_2 & b_2 \\ v_3 & w_3 & b_3 \end{array}\right] \sim \left[\begin{array}{cc|c} v_1 & w_1 & b_1 \\ v_2 & w_2 & b_2 \\ 0 & 0 & b_1^2 + b_2^2 + b_3^2 \end{array}\right]$$

is inconsistent. Therefore $\mathbf{b}$ is not a linear combination of $\mathbf{v}$ and $\mathbf{w}$, as claimed. □

Remark. Note that if we recognize equations in the system

$$\begin{aligned} b_1 v_1 + b_2 v_2 + b_3 v_3 &= 0 \\ b_1 w_1 + b_2 w_2 + b_3 w_3 &= 0 \end{aligned}$$

as the dot products

$$\begin{aligned} \mathbf{v} \cdot \mathbf{b} &= 0 \\ \mathbf{w} \cdot \mathbf{b} &= 0 \end{aligned}$$

then we can see that $\mathbf{b}$ is orthogonal to both $\mathbf{v}$ and $\mathbf{w}$, so geometrically the reason why we choose $\mathbf{b}$ as a non-zero solution to this system should be clear. This says that any non-zero solution to this system is a scalar multiple of $\mathbf{v} \times \mathbf{w}$ (or $\mathbf{w} \times \mathbf{v}$) and, in particular, the solution space is the line in $\mathbb{R}^3$

$$t\left(\mathbf{v} \times \mathbf{w}\right), \, t \in \mathbb{R}.$$

As a concrete example, take the vectors

$$\mathbf{v} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}.$$

We can form the cross-product

$$\mathbf{v} \times \mathbf{w} = \begin{bmatrix} v_2 w_3 - w_2 v_3 \\ -(v_1 w_3 - w_1 v_3) \\ v_1 w_2 - w_1 v_2 \end{bmatrix} = \begin{bmatrix} 2(6) - 3(5) \\ -(1(6) - 3(4)) \\ 1(5) - 4(2) \end{bmatrix} = \begin{bmatrix} -3 \\ 6 \\ -3 \end{bmatrix} = -3 \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

If we let

$$\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

then we can see the equality of the lines parameterized by $t \in \mathbb{R}$

$$t\mathbf{b} = t(\mathbf{v} \times \mathbf{w})$$

and we claim this is the solution space of the system

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 4 & 5 & 6 & 0 \end{array}\right]$$

Indeed, applying the row operations

1. $R_2 - 4R_2$,
2. $-1/3R_2$, and
3. $R_1 - 2R_2$

we obtain the reduced row echelon form

$$\left[\begin{array}{rrr|r} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 0 \end{array}\right]$$

which specifies a family of vectors

$$\begin{bmatrix} t \\ -2t \\ t \end{bmatrix} = t \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = t\mathbf{b}$$

parameterized by $t \in \mathbb{R}$ that are orthogonal to both $\mathbf{v}$ and $\mathbf{w}$.

To see explicitly that $\mathbf{b}$ cannot be expressed as a linear combination of $\mathbf{v}$ and $\mathbf{w}$, observe that

$$b_1^2 + b_2^2 + b_3^2 = 1^2 + (-2)^2 + 1^2 = 6$$

then apply the row operations

1. $b_3 R_3 = 1 \cdot R_3$,
2. $R_3 + b_1 R_1 = R_3 + R_1$, and
3. $R_3 + b_2 R_2 = R_3 - 2R_2$

to obtain the row equivalence

$$\left[\begin{array}{rr|r} 1 & 4 & t \\ 2 & 5 & -2\,t \\ 3 & 6 & t \end{array}\right] \sim \left[\begin{array}{rr|r} 1 & 4 & t \\ 2 & 5 & -2t \\ 0 & 0 & 6t \end{array}\right]$$

which is inconsistent for all $t \neq 0$.